

$$8.2. \quad A^5 + 2A^3 + A = 0$$

$$8.1. \quad A = \begin{pmatrix} -1 & -2 & 0 \\ -1 & 3 & -1 \\ 2 & 3 & 1 \end{pmatrix} \quad f_A(V) \subseteq V$$

$$\det \begin{pmatrix} -1-\lambda & -2 & 0 \\ -1 & 3-\lambda & -1 \\ 2 & 3 & 1-\lambda \end{pmatrix} = (-1-\lambda) \cdot \det \begin{pmatrix} 3-\lambda & -1 \\ 3 & 1-\lambda \end{pmatrix} - (-2) \det \begin{pmatrix} -1 & -1 \\ 2 & 1-\lambda \end{pmatrix}$$

$$(-1-\lambda)(\lambda^2 - 4\lambda + 6) + 2(\lambda + 1) = (\lambda + 1)(-\lambda^2 + 4\lambda - 6 + 2) =$$

$$-(\lambda + 1)(\lambda - 2)^2$$

$$\downarrow_{-1} \quad \downarrow_2$$

$$\lambda_1 = -1$$

$$(A + I)v = 0 \Rightarrow v_1 = (1, 0, -1)$$

$$\lambda_2 = 2$$

$$(A - 2I)v = 0 \Rightarrow v_2 = (2, 3, 5)$$

protože geometrická násobnost je menší než algebraická, hledáme v_3

$$\text{jako řešení } (A - 2I)v_3 = v_2$$

$$\left(\begin{array}{ccc|c} -3 & -2 & 0 & -2 \\ -1 & 1 & -1 & 3 \\ 2 & 3 & -1 & 5 \end{array} \right) \dots \quad v_3 = \left(-\frac{8}{5}, \frac{7}{5}, 0 \right)$$

$$U_1 = \text{Span}(v_1, v_2)$$

$$B_{U_1} = \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} \right\}$$

$$U_2 = \text{Span}(v_2, v_3)$$

$$B_{U_2} = \left\{ \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix}, \begin{pmatrix} -8/5 \\ 7/5 \\ 0 \end{pmatrix} \right\}$$

$$\text{protože } f_A(U) \subseteq U$$